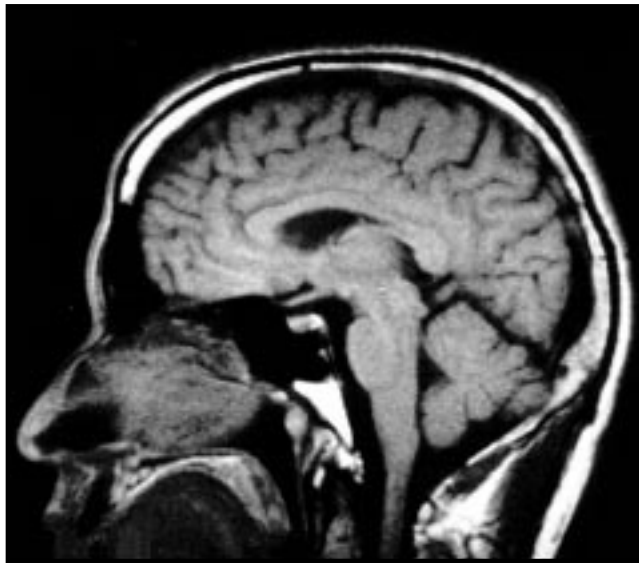


AN8102

9 k W R F A M P L I F I E R



INSTALLATION MANUAL

ANALOGIC ■
*The World Resource
for Precision Signal Technology*

EC Declaration of Conformity

We, Analogic Corporation
Centennial Industrial Park
8 Centennial Drive
Peabody, MA 01960

declare under sole responsibility that the: AN8102 RF Amplifier, to which this declaration relates, meets the essential health and safety requirements and is in conformity with the relevant EC Directives listed below using the relevant section of the following EC standards and other normative documents;

EC EMC Directive 89/336/EEC

Essential health and safety requirements relating to electromagnetic compatibility.

EN55011 Group 2, Class A

Limits and methods of measurements of radio disturbance characteristics of industrial, scientific and medical equipment (ISM) radio frequency

IEC 801-2 Electrostatic discharge requirements for industrial process, measurement and control equipment

IEC 801-3 Radiated electromagnetic field requirements for industrial process, measurement and control equipment

IEC 801-4 Electrically fast transients for industrial process, measurement and control equipment

IEC 801-5 Testing and measurement techniques - Section 5: Surge Immunity

EN60601-1-2 Collateral Standard: Electromagnetic compatibility - Requirements and tests.

EC Medical Device Directive 93/42/EEC

Essential health and safety requirements relating to electrical medical equipment (Medical Devices).

EN 60601-1 Medical Electrical Equipment

IEC 601-1 Part 1: General requirements for safety

EC Declaration 8102 continued

The Product complies with all safety relevant provisions referring to: Protection against electrical hazards
Protection against other hazards such as: mechanical hazards, fire hazards, noise and vibration.

The safety issues of this measurement equipment type have been evaluated by a government accredited
European third party organization such as Intertek Testing Services/ETL.

James R. Levani:  , Corporate Manager Regulatory Affairs

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1. INTRODUCTION

1.1 Safety Precautions

The following safety precautions should be observed while operating this equipment. Failure to comply with these precautions violates safety standards of design, manufacture.

1.1.1 Do Not Open the Cabinet for any Reason

Hazardous high voltages are present inside the cabinet of this instrument. Making contact with any such voltage levels could cause injury or even death. Servicing should be performed only by qualified personnel.

1.1.2 Observe Special Symbols Marked on Equipment

Refer to the instruction manual. The equipment is marked with this symbol when the user is required to refer to the manual for special operating instructions that may prevent any hazard or damage to the instrument.

1.1.3 Observe all Warnings, Cautions, and Notes

The operating instructions in the user's manual include special warnings, cautions, and notes pertaining to the use of this equipment.

1.1.4 Warnings and Cautions

This section is a consolidation of all warnings and cautions used throughout the manual. It is to be used as a reference section only.

WARNING



Hazardous voltages exist in the AN8102 RFPA. It is essential that all operating and repair personnel read and understand this chapter prior to operating the system.

AVERTISSEMENT

De très hautes tensions existent dans le AN8102 RFPA. Il est impératif que tout personnel concerne par la réparation et la maintenance de ce system lisent et comprennent ce chapitre avant la mise en marche.

WARNUNG

Im AN8102 RFPA treten lebensgefährliche spannungen auf. Es ist deshalb unbedingt erforderlich, dass sich debienpersonal und kundenienst vor inbetriebnahme des systems mi dem inhalt dieses kemitels gründlich vertraut machen.

WARNING



The AN8102 contains high voltages. The system should be disconnected from the main power and all high voltage locations discharged before performing maintenance inside the chassis.

AVERTISSEMENT

Le AN8102 contient de très hautes tensions, avant de faire aucune maintenance a l'interieur du systeme, it faut le debrancher de la source principale puis decharger toutes les surfaces a hautes tensions.

WARNUNG

Im AN8102 treten hohe spannungen auf. Vor der durchführung von wartungsarbeiten im gen-hauseinneren sollten der netzanschluss getrennt und alle hochspannungspunkte entladen werden.

1.1.5 System Markings

The following warnings, cautions, labels and silk-screening appear on the Rear Panel of the AN8102 system.

CAUTION

POWER MUST BE OFF
5 MINUTES MINIMUM
PRIOR TO REMOVAL
OF THIS
PANEL

ACHTUNG

ABNAHME DIESER ABDECKUNG
FRÜHESTENS 5 MINUTEN
NACH ABSCHALTEN
DER
NETZPANNUNG

ATTENTION

COUPER L'ALIMENTATION
AU MOURS 5 MINUTES
AVANT D'ENLEVER
CE PANNEAU

PROTECTIVE EARTH



ATTENTION, CONSULT
ACCOMPANYING DOCUMENTS



DANGEROUS VOLTAGE



1.1.6 EMC Compliance

This unit has been tested and found to comply with EN 60601-1-2, EN 55011, Group 2, Class A and FCC Part 18, Subpart C requirements.

Note: Changes or modifications to this unit not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

The following warnings, cautions, labels and silk-screening appear on the Front Panel of the AN8102 system.

CAUTION

POWER MUST BE OFF
5 MINUTES MINIMUM
PRIOR TO REMOVAL
OF THIS
PANEL

ACHTUNG

ABNAHME DIESER ABDECKUNG
FRÜHESTENS 5 MINUTEN
NACH ABSCHALTEN
DER
NETZPANNUNG

ATTENTION

COUPER L'ALIMENTATION
AU MOURS 5 MINUTES
AVANT D'ENLEVER
CE PANNEAU

WARNING

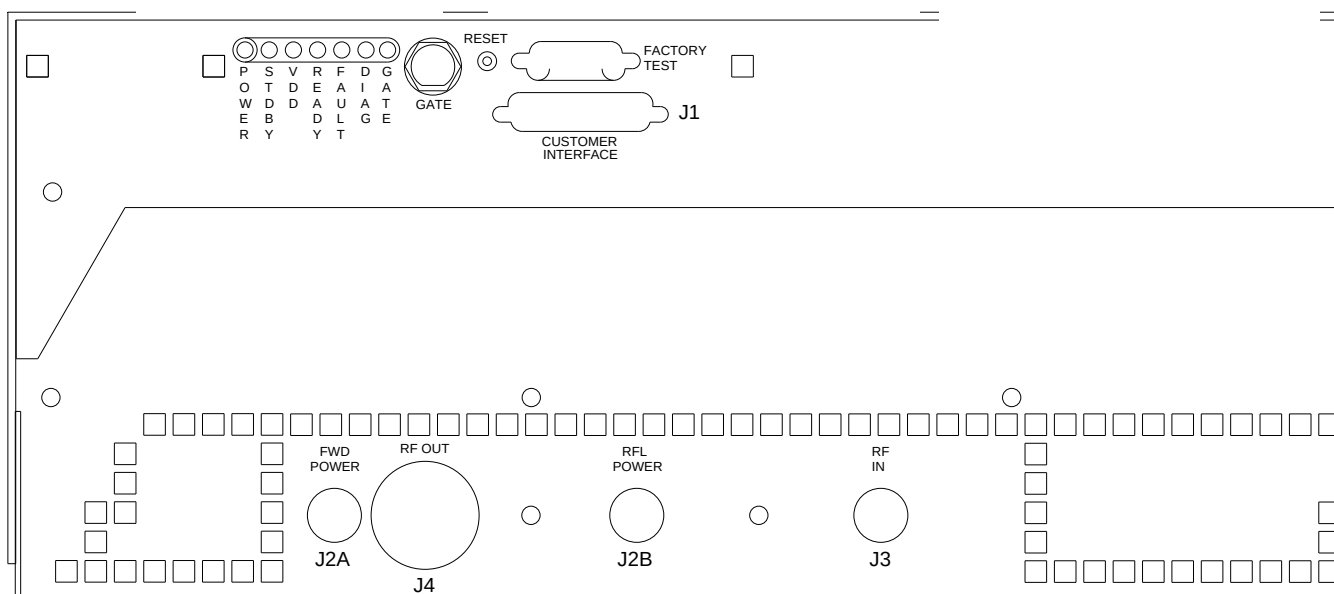
ASSEMBLY WEIGHT IS
33 kg. TWO PERSONS OR
A LIFTING FIXTURE ARE
REQUIRED FOR REMOVAL

AVERTISSEMENT

L'ENSEMBLE PESE 33 kg.
IL NE DOIT ETRE RETIRE QUE PAR
DEUX PERSONNES
OU AU MOYEN D'UN
DISPOSITIF DE LEVAGE

WARNUNG

DAS GEWICHT DER BAUGRUPPE
IST 33 kg.
ZUM ENTFERNEN MUSSEN
ZWEI PERSONEN ODER
EINE HEBEVORRICHTUNG
ZUR VERFUGUNG STEHEN



The following warnings, cautions, labels and silk-screening appear on the Side Panel of the AN8102 system.

**ANALOGIC
PEABODY, MASS.**

MOD.

S.N.

V~

A

Hz

OUTPUT W

1.2 Packing List and Unpacking Instructions

The AN8102 is shipped in a single crate containing the following items:

- ☐ AN8102 System Chassis.
- ☐ The following documentation:
 - AN8102 Installation Manual (this manual)

Remove all items from their packaging.

Note: If your shipped package does not contain all items listed above, contact your Analogic sales or support representative at (978) 977-3000.

1.3 General Description

The AN8102 is a stand-alone RF power amplifier intended for use in a Magnetic Resonance Imaging (MRI) system. Amplifier connections and indicators are straightforward for easy operation.

All I/O connectors, controls and indicators are placed on the rear panel as shown in Figure 1-1.

- ☐ System status and diagnostic LEDs.
- ☐ HN-type RF Output connector.
- ☐ BNC RF Input connector.
- ☐ BNC connector for forward power sample.
- ☐ BNC connector for reflected power sample.
- ☐ BNC connector for gating.
- ☐ Fault/overload RESET button.
- ☐ Connector for communication between the remote controller (MRI system) and the AN8102.
- ☐ Customer specific connector for communication.
- ☐ M5 terminal block for AC input power.
- ☐ AC power circuit breaker.

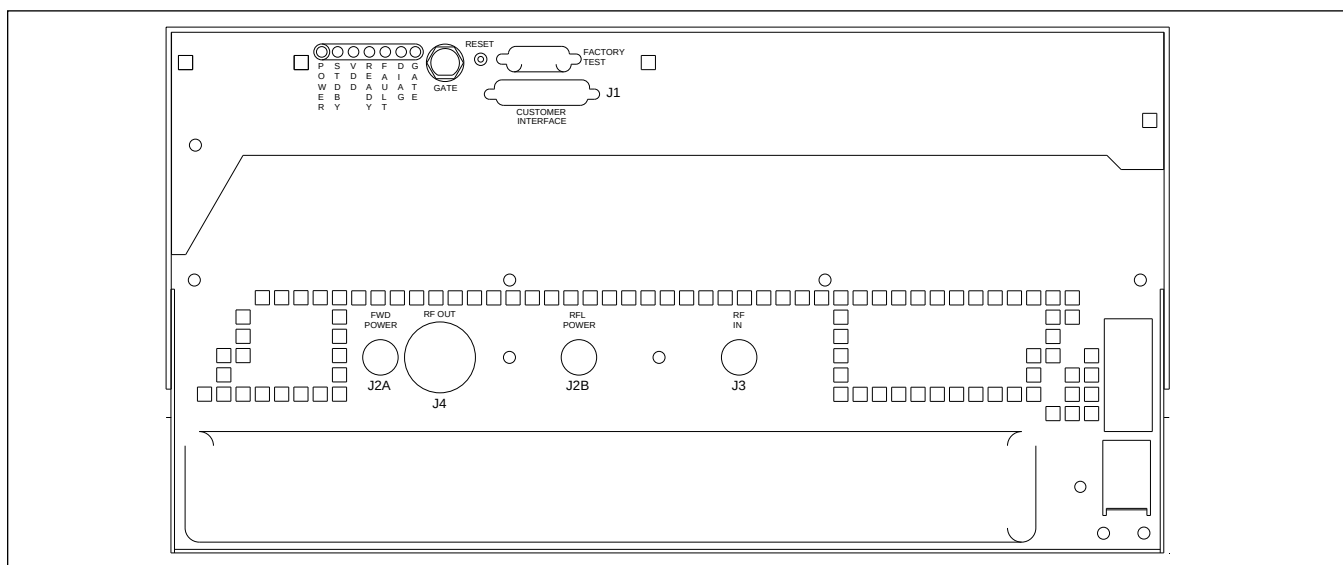


Figure 1-1. AN8102 Rear Panel.

The fans are mounted on the front panel as shown in Figure 1-2.

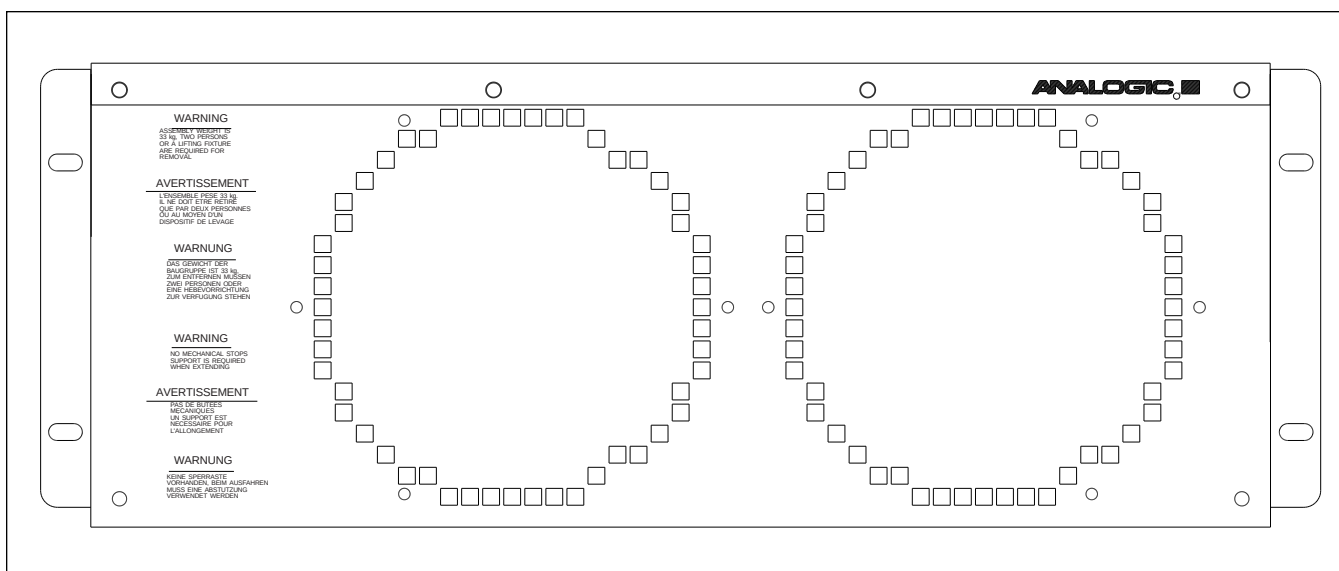


Figure 1-2. AN8102 Front Panel.

1.4 Installation

After unpacking the Amplifier, it may be table mounted for testing and then may be placed in a standard 19 inch electronic instrument rack.

When locating the unit in an enclosure, be sure that the front and rear cooling vents and fans are unobstructed to ambient room air. Blockage of these vents may cause overheating of the Amplifier, activation of a status protection system, or premature failure of components.

Refer to Section 4 for system operation information.

NOTE

This unit is specified to operate in a non-condensing atmosphere. If it is allowed to sit in a condensing environment, for example, non-operating in an uncontrolled humid room, then it must be allowed to dry out in a humidity-controlled room prior to operation.

2. SPECIFICATIONS

The AN8102 is a high power electronic device and must be operated within its specified limits. Although the Amplifier has a number of built-in fault and overload detectors, care should always be taken that design parameters are strictly adhered to when developing your MRI system.

2.1 Frequency Characteristics

- | | |
|------------------------------|---|
| 1. Operating Frequency range | 63.86 MHz hardware preset nominal center frequency, ± 500 kHz |
| 2. Gain Variation | max.: ± 0.2 dB over ± 275 kHz around the center frequency |

2.2 Large Signal Characteristics

- | | |
|------------------------|---|
| 1. Output Power | (into 50Ω load) |
| P_{peak} | 9.0 kW nominal (69.54 dBm) = P_{nom} |
| P_{average} | 900W @ 9.0 kW (PEP) |
| | $900 \times \sqrt{\frac{P_{\text{out}}}{9000}} \quad \text{for } P_{\text{out}} \leq P_{\text{nom}}$ |
| 2. Gain | $10 \times \log(P_{\text{nom}}) + 30 \pm 0.5$ dB (Measured at full output power) |
| 3. RF Action | $T^2 \times P_{\text{peak}} = 225$ for a single pulse where T = Pulse Width in ms and P_{peak} = Peak Envelope Power in kW |
| 4. Amplitude Linearity | 2 dB Max. over any 40 dB dynamic range up to full output |
| | IMD less than -20 dBc (measured similar to MIL-STD-1311 & 2204B; frequency difference: 10 kHz) |
| 5. Phase Linearity | Max.: 10 degrees over any 40 dB dynamic range up to full output |
| 6. Matching | Input impedance 50Ω nominal |
| | Input VSWR 1.5 Maximum |
| | Output impedance 50Ω nominal |
| | Output VSWR 4.0 Maximum |
| | Power delivered to load $\frac{P_{\text{nom}}}{\text{VSWR}}$ |
| | Load is a tuned coil system with nominal 50 Ω resistive at the center frequency |
| | Amplifier is operational and unconditionally stable for any passive load |

2.3 Output Power Stability

1. Long term stability
 ± 0.5 dB/3000 Hr.
2. Short term stability
Max.: 0.15 dB gain variation
Max.: 2 degrees phase variation
(Within 5 minutes at constant input level after 20 minute warm-up time with:
 $P_{\text{peak}} = P_{\text{nom}}, P_{\text{average}} = 10\% * P_{\text{nom}}$
Pulse Width = 5.0 ms and a 10 minute break without RF and gating)

2.4 Gating Characteristics

1. Gating Duty Cycle
50% Maximum
2. Pulse Width T_{max}
60 ms Maximum (P_{peak} is derated based on RF Action specification)
3. Gain Variation during the pulse
(Pulse width governed by RF Action and for pulse width ≤ 30.0 ms)
0.4 dB maximum (Measurement to begin after 1% of pulse width or 50 μs whichever is greater with a 100 kHz bandpass filter)
4. Gain Variation during the pulse
(Pulse width governed by RF Action and for 30 ms $\leq$ pulse width $\leq T_{\text{max}}$)
Maximum ± 1.0 dB gain variation after 1% of pulse width, or 50 μs (whichever is greater), with a 100 kHz bandpass filter
5. Gate switching transitions
Gating ON time 10 μs Max.
(0 to 90% of V_{nom})
Gating OFF time 5 μs Max.
(100% to 10% of V_{nom})
Gating OFF time 10 μs Max.
(to blanked noise level)

2.5 Spurious Signals and Noise

1. Harmonic Distortion
<-30 dBc all components
2. Periodic ripple and Spurious
Larger of -60 dBc or -20 dBm
3. Noise spectrum
<-160 dBm/Hz @ 25°C ambient, Gated off
<-85 dBm/Hz, Gated on with RF Input terminated with 50 Ω

2.6 Safety of Operation

1. Self Protection
Protection against input overload of maximum +20.0 dBm
2. Safety Requirements
Even with one defective device, neither input nor output may carry any DC voltage
Safety against any combination of input and output conditions within allowable ratings

2.7 Regulatory Requirements

The unit is certified by a NRTL to meet the following regulations as a system component:

IEC 601-1, IEC 601-1-2 ed. 2

UL2601-1

CSA C22.2 #601.1-M90

FCC Part 18 Class A, and Part 15

EN 55011 Class A Group 2 (CISPR 11)

Description:

Model AN8102 Radio Frequency Amplifier

Manufactured by Analogic Corporation

8 Centennial Drive

Peabody, MA 01960 USA

Classification: Class 1 medical equipment, IPXX ingress rating, Continuous operation,

- No applied parts.
- No fuses or internal parts replaceable by user.
- Filters to be removed for cleaning or replacement; only dry filters to be installed only by authorized service technician.

Symbols on unit:



Electrical shock hazard — High voltages present on exposed terminals



Protective earth terminal



Caution: Power must be off 5 minutes minimum prior to removal of this panel



Functional earth ground stud



Equipotential stud

2.8 Reliability

MTBF: 25,000 Hr. (Calculated according to MIL-HDBK-217F)

2.9 Interface Connections

1. RF Monitoring Output	RF – monitoring output for Forward and Reflected signals	
	Forward Coupling	-30 ±0.5 dB
	Directivity	>20 dB
2. RF Connectors	Input connector	BNC female
	Output connector	HN female
	RF monitoring	2 X BNC female
3. Control connector (Rear panel)	25 pin “D” Female with No 4-40VNC-2B female mounting hardware. (See below for signals and pin numbers)	
4. Gating connector (Rear panel)	Not applicable	
5. Rear panel indicators & switches	POWER (Green LED)	
	STDBY (Yellow LED)	
	VDD (Yellow LED)	
	READY (Yellow LED)	
	FAULT (Yellow LED)	
	DIAG (Yellow LED)	
	GATE (Yellow LED)	
	FAULT/OVERLOAD RESET SWITCH	
6. Rear panel AC connector & switches	Circuit Breaker for AC Mains	
	Field Wiring Terminals for AC Mains Input: TB with M5 ground stud	
7. AC Mains	180 to 253 VAC; 1 ϕ	
	47 – 63 Hz	
Power Consumption	<4000 VA	
Inrush current during turn on	<2 X Nominal current	

2.10 Mechanical and Environmental

1. Mechanical Dimensions	19 inch EIA Rack mountable	
	Maximum Height	178 mm
	Maximum Depth	686 mm
	Maximum Weight	36 kg
2. Front Panel finish	Brushed Aluminum, Yellow Chromate	

Non-operating Storage Temperature test, to be performed in the shipping container:

Dry heat: IEC68-2-2(Bb) 70°C for 96 hours

Dry cold: IEC68-2-1(Ab) -25°C for 72 hours

Damp heat: IEC68-2-3(Ca) 40°C, 93% (+3/-2%)RH, non-condensing for 96 hours. Test data to be taken between 1 and 2 hours after removal from damp heat.

Operational temperature: The unit shall be designed to operate continuously under the following environmental conditions, and tested as follows:

Cold: IEC68-2-1(Ad) 10 °C, 20% to 50% RH for 16 hours
 Dry heat: IEC68-2-2(Bd) 40 °C, 20% to 50% RH for 16 hours
 Steady state damp heat: IEC68-2-3 (Ca) 40°C, 85% RH non-condensing for 96 hours.
 Test data to be taken 2 hours after each temperature stabilization.

Altitude: 2500 meters maximum operating and non-operating

Magnetic field: 5 mT maximum operating and non-operating

Mechanical bump and vibration: (Unit shall pass without damage):

Bump: IEC68-2-29(Eb): 10 G's, 6-10 ms pulse duration, 1000 bumps in each of 3 axes. Each chassis non-operating in normal operating position, held to the flat fixture with rods attached to top bars over unit.

Vibration: IEC68-2-6(Fc) <100 lbs; 2 G's, 10 Hz-150 Hz-10 Hz, 1 octave/min.

>100 lbs; 2 G's, 10 Hz-55 Hz-10 Hz, 1 octave/min.

Each chassis non-operating in normal operating position, held to the flat fixture with rods attached to top bars over unit.

Transportation and shipping containers: National Safety Transit Assoc. Project 2A for packaged products under 100 lb when prepared for shipment, and Project 2 for packaged products over 100 lbs when prepared for shipment.

2.11 Customer Interface Connector Signals and Pin Numbers

Signal Name	Pin Number	Interface Type
Input Signals from Customer		
Not Unblank 0° amp +	1	RS-422
Not Unblank 0° amp –	14	RS-422
Not Unblank 90° amp +	2	RS-422
Not Unblank 90° amp –	15	RS-422
Not Main power 1 +	3	RS-422
Not Main power 1 –	16	RS-422
Not Main power 2 +	4	RS-422
Not Main power 2 –	17	RS-422
Not Reset +	5	RS-422
Not Reset –	18	RS-422
Output Signals to Customer		
Not connect +	6	Connect to 19
Not connect RTN	19	Connect to 6
Status 5 MSB +	7	RS-422
Status 5 MSB –	20	RS-422
Status 4 +	8	RS-422
Status 4 –	21	RS-422
Status 3 +	9	RS-422
Status 3 –	22	RS-422
Status 2 +	10	RS-422
Status 2 –	23	RS-422
Status 1 +	11	RS-422
Status 1 –	24	RS-422
Status 0 LSB +	12	RS-422
Status 0 LSB –	25	RS-422
Ground	13	GND

Digital Status Signal Bit Definition Table

Status Bit msb lsb	Status Code	LEDs Extinguished indicate ● Lit indicate ○				Description
		V v d	R e a d y	S t a n d b y	F a u l t	
000000	Hex 00	●	●	●	●	No power to RFM. RFM interface not active.
000001	Hex 01	○	○	●	●	Operation State. Ready for RF.
000010	Hex 02	○	●	○	●	Transient State (Preparing for operation) Follows Main_Power1&2
000011	Hex 03	●	●	○	●	Low Power Dissipation – System Idle
000100 ≡ 010111	Hex 04 ≡ Hex 17					Not Used
011000 ≡ 101111	Hex 18 ≡ Hex 2F					
		—	●	●	○	Fault codes listed below

The following table describes the fault codes for the AN8102G RFA. The hex and binary coding is referred to the 6-bit AN8102G status interface.

The LED Flash Code refers to the sequence flashed on the DIAG LED.

Voltage levels, low or high, refer to the internal power supply voltages. Device fault refers to the RFA FET device and its surrounding circuit. PSU refers to the high voltage power supply, a combination of either internal over current (blown FET) or heat sink temperature. Duty cycle refers to the gate duty cycle. I peak refers to instantaneous current to the RF amp section. Control EE refers to the EEPROM in the microprocessor, diagnosed by checksum. ADC refers to the internal error detection in the RFA ADC. Heat sink temp refers to the main RFA heat sink. Peak, average, and reflected power levels refer to the output of the directional coupler. Gate width refers to the maximum time width of the gate pulse. T junction refers to the real-time computed RFA FET junction temperature.

FAULT/OVLD Serial link message	STATUS CODE Hex	STATUS BIT Binary	LED Flash Code
"-15V Low"	Hex 18	01 1000	3-5
"-15V High"	Hex 19	01 1001	5-1
"15V Low"	Hex 1A	01 1010	3-4
"15V High"	Hex 1B	01 1011	4-6
"48V Low"	Hex 1C	01 1100	3-3
"48V High"	Hex 1D	01 1101	4-5
"HV Low"	Hex 1E	01 1110	3-2
"HV High"	Hex 1F	01 1111	4-4
"Device Fault"	Hex 20	10 0000	1-2
"PSU"	Hex 21	10 0001	1-4
"Duty Cycle"	Hex 22	10 0010	1-3
"I Peak OVLD"	Hex 23	10 0011	2-2
"Not used" (DeltaTempHi)	None	None	4-1
"Ctrl EE err"	Hex 25	10 0101	5-2
"ADC Err"	Hex 26	10 0110	5-3
"Not Used" (CPU)	None	None	2-3
"Not Used" (Drive)	Hex 28	10 1000	None
"HeatSinkTmpHigh"	Hex 29	10 1001	3-6
"Peak Power"	Hex 2A	10 1010	1-6
"AveragePowerHigh"	Hex 2B	10 1011	4-2
"Reflection Power"	Hex 2C	10 1100	2-1
"GATE Width Long"	Hex 2D	10 1101	2-4
"TJunction Fault"	Hex 2E	10 1110	1-5
"Unknown"	Hex 2F	10 1111	None
"Bias Low"	None	None	3-1
"Bias High"	None	None	4-3

2.12 Factory Interface Control Connector

Signal Name	Pin Number	Interface Type
Input Signals from Customer		
AC ON +	1	RS-422
AC ON –	9	RS-422
FAULT RESET +	2	RS-422
FAULT RESET –	10	RS-422
Output Signals to Customer		
AMP READY +	3	RS-422
AMP READY –	11	RS-422
AMP FAULT +	4	RS-422
AMP FAULT –	12	RS-422
AMP OVERLOAD +	5	RS-422
AMP OVERLOAD –	13	RS-422
RS-232 Communication Signals		
DATA FROM AMPLIFIER (OUTPUT)	6	RS-232-RxD
DATA TO AMPLIFIER (INPUT)	7	RS-232-TxD
GROUND	15	
NO CONNECTIONS	8, 14	

2.13 Software Operation

2.13.1 Communications Protocol for Factory Test Port

The amplifier supports a serial communication link, allowing to perform diagnostics, control the operation of the amplifier and set options. The communications interface protocol is command line oriented. Any ASCII terminal, or a compatible device, such as a computer running a generic terminal-emulation program, may be used as a host.

Communications link is pre-configured to run at 9600 baud, 8 data bits, 1 stop bit, no parity and no flow control (Xon, Xoff). To reset the communications to this default setting:

1. Turn power on subsystem.
2. Push Fault RESET button on the front panel.
3. Send RESET command via customers link.

2.13.1.1 Command Line

The interface provides full command line editing capability.

Key	ASCII Decimal Code	Meaning
<Backspace>or<BS>	8	Move cursor to the left.
<Delete>	127	Delete the (invisible) character.
<Return>or<CR>	13 (n)	Execute command line.
<Linefeed>or<LF>	19 (r)	Execute command line.

In the following descriptions, “\n” will be used to denote a return or pressing the enter key. Every command consists of a keyword and a list of parameters, delimited by any number of spaces. Also, any number of spaces are allowed anywhere except inside words and numbers. Command line first two letters are case sensitive (**MODE** and **Mode** and **MO** are acceptable, **mode** is not). Sometimes command can consist of a few first characters highlighted by upper case letters (See command descriptions). Error message “Bad keyword” is returned in any other case. Whenever a number is required as a parameter, it can be entered in decimal, hexadecimal (prefixed with “0x”), or octal (prefixed with “0”) notation, for example, 123 (Decimal) = 0x7B (Hex) = 0173 (Octal). The maximum size of the command line is 40 characters.

All parameters must be unsigned integers and have a defined range of values. Values beyond the defined range may give a wrong result.

Examples of erroneous input:

-missing space between [?] and the first digit returns an error message:

“Argument(s) required.”

-number DAC more than max. number (39) returns an error message:

“Error during execution.”

-erroneous argument such as missing 0x in hex value 0xFF, or using signs, instead inputting FF, -66, +66, or F1 will result in a response of “0” if a DAC number is expected, or will result in no response if the parameter required is a location such as EE:RF;

-erroneous responses may be truncated, such as 1F is interpreted as 1;

-0xFFFF is interpreted as 255;

-value more than range allowed will result in a FAULT message.

2.13.1.2 Command Description

Note: Systems with only one RF module do not support SLAVE commands.

HELP?

HELP prints a list of all available commands with short descriptions.

If you missed “?”, system displays error message:

“Only query allowed.”

If you printed wrong command, system displays error message:

“Bad keyword.”

If you printed wrong command’s argument, system displays error message:

“Too many arguments.”

MMode?

Report the operational status.

Status	What it means
Start	System microcontroller initializes software and hardware after power on.
Warmup	The amplifier is powering up.
Run	External gating is enabled, amplifier ready.
Bias	main RF power is turned on, measure bias.
Off	Main RF power off, gating disabled.
Stopped	The amplifier is powering down.
Overload: <overload name>	Amplifier is overloaded, <overload name> indicates the type of overload.
FAULT: <fault name>	Amplifier is faulted, <fault name> indicates the type of fault.
Recover	System re-initializes software and hardware after fault or overload state and RESET

RESet \n

Reset the amplifier from overload or fault. If the reset succeeds, **MMode** will change to **Recover** and then, in 5 seconds maximum, to **Off**.

Off \n

Disable both the gating and the main power. If the amplifier is already turned on by remote or if Autostart jumper is installed, the system will return the message “**Invalid Mode.**”

ON \n

Enable the gating and the main power. If the amplifier is already turned on by remote or if Autostart jumper is installed, the system will return the message “**Invalid Mode.**”

CONF: <setting>?

CONF: <setting> <new value \n

Display or change set-up data. Any changes made to the settings will be in effect until the amplifier is powered down.

Keyword for <setting>	Possible values for <new value>	What it does
DEV1	EN or Dis	Use DI (disable Device 1 Fault), to permit operation with a single output stage device failure.
AUTOMat	EN or Dis	Enable the automatic overload reset feature in 15 seconds after an overload has occurred.
BROADcast	EN or Dis	Enable immediately display fault or overload message un-solicitedly.
TERMinal	EN or Dis	Enable display echo of commands

IDENTify

The identify command is used for assignment of SERIAL number and REVISION of the CONTROL board (by default) or the RF board. Both the control board and RF are MASTER by default because they connect with the terminal via D-connector. SLAVES connect only via IPS bus. If you print :SLAVE for the argument of this command, the system displays error message:

Slaves are not supported in this version.

Commands for control board look like:

IDEN:CTRL:SER <String req'd> \n (password protected)

for example **IDEN:CTRL:SER COO4L119 \n**

IDEN:CTRL:REV <String req'd> \n (password protected)

for example **IDEN:CTRL:REV 01C \n**

String requested a number revision or version control board or RF.

Commands for RF look similar:

IDEN:RF:SER <String req'd> \n (password protected)

for example **IDEN:RF:SER COO49Q7 \n**

IDEN:RF:REV <String req'd> \n (password protected)

for example **IDEN:REV 02H \n**

This command can also be used as a query. For example,

IDEN:RF:SER? will return the serial number of RF Printed Circuit Board and

IDEN:CTRL:REV? will return the Revision number of Control/PSU Board.

FAULT?

An overload is an error condition that is likely to occur due to improper use of the amplifier. If an overload or a fault is detected, both the High Voltage and the external gating are automatically disabled. A fault is an error condition that is likely to occur due to an internal circuit or component failure. However, in case of a power supply current limit exceeded due to excess duty cycle demand, the system will always respond **Fault: HV Lo**. If a fault is detected, the amplifier automatically shuts off. For overload and fault conditions, when detected, the amplifier will send a message **OVERLOAD** or **FAULT** with appropriate code depending on the condition. The user has to inquire to determine the actual overload or fault condition by using the **FAULT?** command. In case of an overload, the system can automatically reset itself every 15 seconds until an external reset is asserted either through the RESET line or through a **RESET** command from the host. This automatic reset feature is enabled through **CONF:AUTO EN** command. The system will be ready only after the overload condition is removed. If fault or overload exists, the system displays the appropriate message.

BIAS:TARG <value> \n (password protected)

Assign value (100-510mA) for bias; entry in mA.

BIAS?

Check and measure current bias. System displays current value in mA.

BIAS:AUTOMATIC (password protected)

Adjustment of bias current is automatically done by the amplifier, as would be done manually with the **BIAS:TARGET** value.

Check device bias. Measure bias current of the output and driver stage devices.

PASS <Password> \n

Assign password to work with several commands: **EE:ERASE**, **BIAS**.

Using these commands without **PASS**, system displays error message:

“Protected command”

Print wrong password, system displays error message:

“Bad keyword.”

MEASURE?

System displays the following parameters in the following order:

- RF heat-sink temperature (°C);
- Temperature rise from RF heat-sink to RF Power FET junction (°C);
- RF Output average power delivered to the load (watts);
- Bias (mA) (Usually shows as “0.0” upon query);
- DC Drain voltage for RF Power FET (volts);
- +28V (volts);
- +15V (volts);
- –15V (volts).

EEPROM

ACCESS, CHECK, ERASE, FIX, LOAD and STORE in EEPROM information (in MASTER unit by default).

EEPROM:RF? <number DAC> \n (password protected)

Read from EEPROM a value DAC, that number you print.

EEPROM:RF <number DAC> <value> \n (password protected)

Write into EEPROM a value DAC, that number you print.

EEPROM:CHECK?**EEPROM:RF:CHECK?****EEPROM:CTRL:CHECK?**

Check checksum CONTROL board or RF data in EEPROM. System is reporting:

EE checksum Bad or Good. After ERASE and STORE commands

checksum should be Good.

EEPROM:ERASE \n (password protected)**EEPROM:ERASE:RF \n** (password protected)**EEPROM:ERASE:CTRL \n** (password protected)

Erase data CONTROL board or RF in EEPROM.

EEPROM:FIX \n (password protected)**EEPROM:FIX:RF \n** (password protected)**EEPROM:FIX:CTRL \n** (password protected)

Fix data CONTROL board or RF in EEPROM.

EEPROM:LOAD \n**EEPROM:LOAD:RF \n****EEPROM:LOAD:CTRL \n**

Load data CONTROL board or RF from EEPROM into RAM.

EEprom:STOre \n
EEprom:STOre:RF \n
EEprom:STOre:CTRL \n

STOre data CONTRol board or RF from RAM into EEPROM and form checksum.

DAC? <number DAC> \n (password protected)
DAC:SLA? <number DAC> \n (password protected)
Read from RAM a value DAC (in MASTer unit by default) with the selected number.

DAC? * \n (password protected)
Read from RAM all values of DAC (in MASTer unit by default).

DAC <number DAC> <value> \n (password protected)
DAC:SLA <number DAC> <value> \n (password protected)

Write into RAM a value in DAC (in MASTer by default) with selected number.

DAC:SLA refers to slave module DAC.

Note: If you print :SLA, system displays error message:

“Bad keyword”

If you print :SLAve, system displays error message:

“Too many arguments”

PROD <pulse width> \n (password protected)

The Product Test command is a special one to generate a test gating pulse pattern. It requires one parameter, the pulse width in milliseconds, which is rounded downward to the next lower multiple of 4 milliseconds (for example, 0, 1, 2, or 3 means 0; 4, 5, 6, or 7 means 4, etc.) The maximum pulse width is 48 milliseconds. The pulse is repeated every 48 milliseconds. The pattern continues until any key is pressed, unless a device fault is detected. Devices are tested for bias current within +12.5% to -37.5% of the target bias current value. The values of bias current are reported every minute during the test. If a bias error is detected, the test is terminated and the Q numbers of the failing FETs are printed. If a key is pressed, the test is terminated and “Test Done” is printed after showing the bias current of the FETs at that time.

2.13.1.3 Communications Interface

The amplifier software supports a customer communications link as described in section 2.11. The status and LED states are described in the digital status signal definition table.

3. INSTALLATION

The AN8102 should be wired into your MRI system in conformity with the information in this section.

3.1 AC Line Connections

The AC Line connections are accessible from the field wiring box on the rear panel of the amplifier. The recommended cordage to use for connecting the system is Belden type SJT cordage Cat. No. 19353 rated at 300V or equivalent.. The system is designed to operate from 200, 208, 220 or 230 VAC. Depending upon where the system is to be installed, the input voltage can be obtained by connecting the system between two phases that measure the rated voltage phase to phase or it can be connected from one phase to neutral. Determine the voltage used at the installation site and connect the system in the configuration that provides one of the rated input voltages. The input connections are made at the input terminal block. See Figure 3-1 for terminal block location. The phase to phase or phase to neutral voltage can be connected to the terminal block in either polarity. The ground wire is connected to the ground stud to the left of the terminal block. **NOTE:** *Make sure the ground lug is mounted closest to the chassis followed by the lock washer nut.* A cable clamp is also provided as strain relief for the line cord.

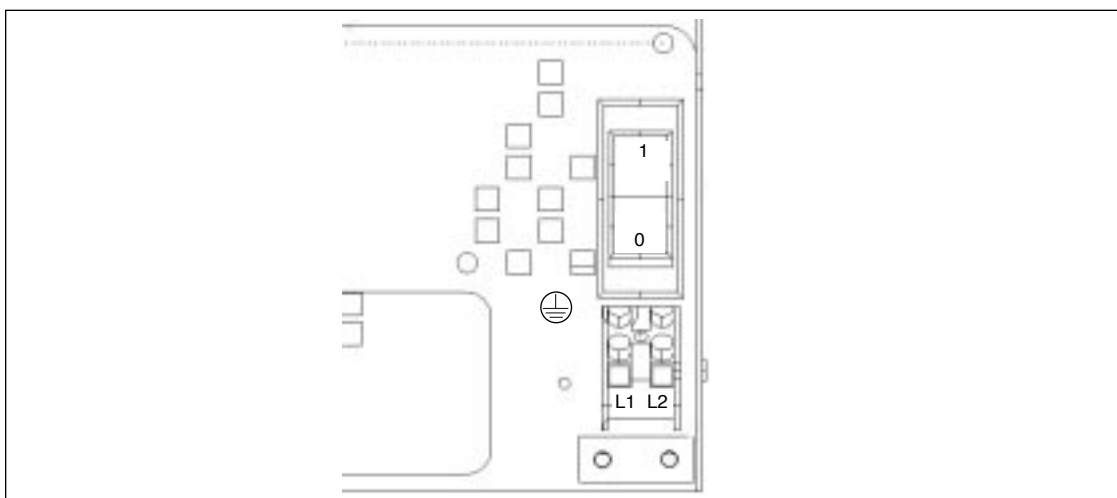


Figure 3-1. AC Line Connections.

3.2 RF Connections

The following table describes the RF signals and connectors used or provided by the AN8102.

Table 3-1. AN8101 RF Interconnections.

Connector	Type	I/O	Description
RF IN	BNC	Input	0 dBm nominal RF input
RF OUT	HN	Output	9.0 kW PEP RF pulse output
GATING	BNC	Input	Pulse gating signal
FORWARD MONITOR	BNC	Output	Forward RF sample -40 $\pm$ 1 dB coupling attenuation
REFLECTED MONITOR	BNC	Output	Reflected RF sample 25 dB directivity referred to -40 dB coupling

3.3 Interface Connections

The following table describes the signals used to control the AN8102. All signal inputs (except RS-232 comm) are differential to internal optoisolators, while the outputs are differential for isolation by the user.

Table 3.2. Customer Interface Signals.

Signal Name	Pin	Asserted	Not Asserted	Function when Asserted
Input Signals				
AC ON +	1	H	L	Turns on AC Power to amplifier.
AC ON -	9	L	H	
FAULT RESET +	2	H	L	Resets system Faults or Overloads
FAULT RESET -	10	L	H	
Output Signals				
AMP READY +	3	H	L	Amplifier Ready
AMP READY -	11	L	H	
AMP FAULT +	4	H	L	Amplifier in Fault mode
AMP FAULT -	12	L	H	
AMP OVERLOAD +	5	H	L	Amplifier in Overload mode
AMP OVERLOAD -	13	L	H	
RS-232 Comm Signals				
TXD	6			Xmit data from Amplifier.
RXD	7			Rcvd data from Host system.
NC	8			No Connection
NC	14			No Connection
GND	15			

The customer-specific interface connector is described in the specification section 2, and is supplied only with the AN8102G model.

4. SYSTEM OPERATION

4.1 Start-Up

Once the AN8102 is properly configured in the user system, the unit may be turned ON. The start-up sequence is as follows:

- ☐ Attach the power cord to the AN8102 and connect to the power outlet as shown in Section 3.1.
- ☐ Make signal connections as discussed in Section 3.2 and 3.3.
- ☐ Turn the System ON by switching the breaker located on the rear panel. The front panel POWER LED will light.
- ☐ Assert AC ON signal to start power-up sequencing. The READY LED will flash during power up then light when the start-up sequence is finished. Note: If the system is configured for AUTO START, the system will start the power-up sequence upon turning the AC mains circuit breaker on. In this mode, there is no need to provide remote AC ON signal at the customer interface connector.
- ☐ The System is now controlled by Remote Interface signals as explained in Section 4.2.

4.2 Operation

A block diagram of the AN8102 is given in Appendix A.

The RF amplifier is controlled through the Customer Interface connector. The interface signals and their active levels were presented in Table 3-2 of Section 3. The general control sequence is as follows:

1. Assert AC ON if necessary.
2. Wait for READY signal to be asserted, indicating that the system is ready for RF output.
3. Connect the GATING signal at the desired gating frequency and duty cycle with appropriate polarity for pulsed RF output.
4. Monitor FAULT and OVERLOAD lines to determine whether the system is in operational condition..

System overloads may be reset by asserting the FAULT RESET line or by pressing the front panel RESET switch. The system will then go to Step 2, above.

4.3 LED Indicators

The front panel (shown in Figure 4-1) contains a bank of LED indicators to inform the operator of the status of the amplifier. The front panel indicators are divided into 2 sections - STATUS and RF GATE. The following sections describe the functions of each LED.

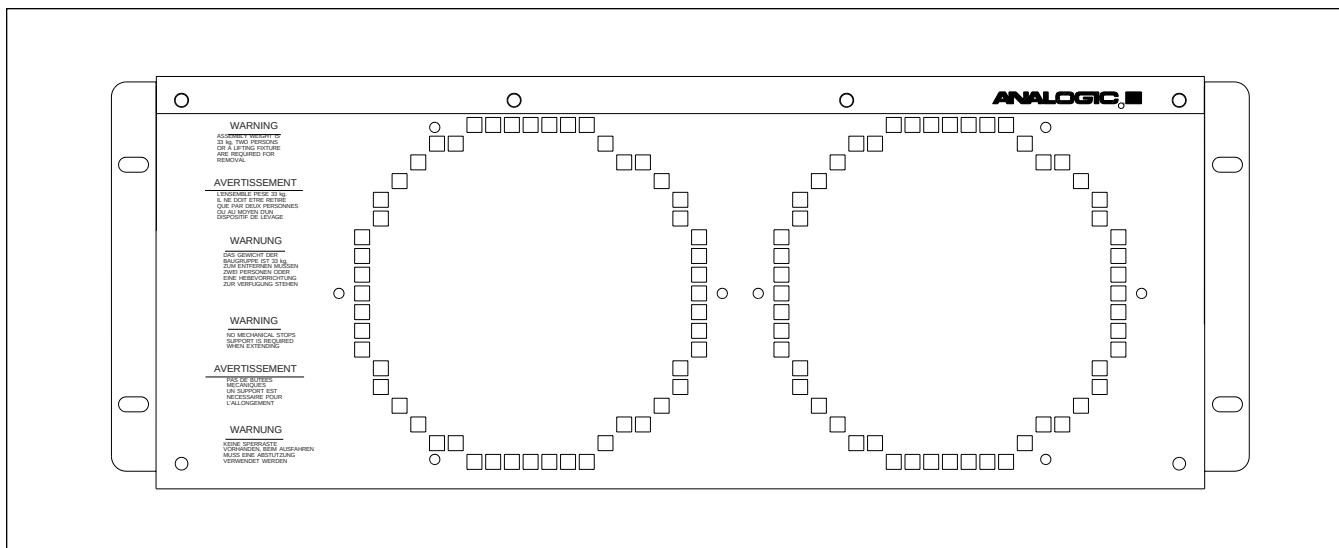


Figure 4-1. Front Panel Indicators and Controls.

4.3.1 Status Indicators

See section 2 (specification); section 2.11 for the STATUS LED descriptions.

4.3.2 Diagnostic Indicator

The READY LED is also used together with the FAULT LED to inform the operator the cause of system shutdown.

4.3.2.1 FAULT / OVERLOAD Indicator

When a system failure occurs, the AN8102 attempts to determine whether it is due to an internal hardware problem or a user condition. This decision is based upon a review of sensor readings and an algorithm stored in programmable logic on the Control Board. The result is that the FAULT LED lights for an internal hardware failure and for most overdrive and load mismatch conditions. In addition, the READY LED flashes to identify the particular FAULT or OVERLOAD. Refer to Table 4-1 for Fault and Overload Codes.

A Fault or overload indication results in the following actions:

1. External Gating is disabled, preventing RF Output.
2. HV is turned off.
3. Attempted resets (manual, external, or automatic) will return the amplifier to READY status if the fault or overload condition has been corrected. The system will continue to FAULT if there is a permanent component failure.

Note: *The system also has a feature of automatic internal reset that will occur at 15 seconds interval for all overload conditions. This feature can be enabled through the RS232C link from the 15-Pin “D” Shell Interface Connector. The instructions for appropriate command syntax can be obtained by sending a “HELP” command through the RS232C link. This feature is DISABLED for all systems shipped from ANALOGIC.*

4.3.3 System Error Codes

The error code is defined in terms of quarter-second time units. The READY LED will flash a THREE-DIGIT code and the number flashes will determine the digits. There will be a 1 second pause between the three digits and the error code will repeat every 3 seconds. The ERROR CODEs are categorized as follows:

The FIRST digits are categorized as

1. Overload
2. Fault

The second and third digit represents the appropriate fault or overload condition.

For example: for a JUNCTION TEMPERATURE OVERLOAD, the READY LED will flash 1 time – 1 second pause, 1 time – 1 second pause, 5 times and this will be repeated every 3 seconds.

Table 4-1. ERROR CODES.

Fault	Flash Code
Device Fault	1 – 2
Duty Cycle Overload	1 – 3
Power Supply Fault	1 – 4
Junction Temp Overload	1 – 5
Peak Forward Power Overload	1 – 6
Peak Reflected Power Overload	2 – 1
Peak Current Overload	2 – 2
Microprocessor Fault	2 – 3
Gate Width Too Long	2 – 4
(Not Used)	2 – 5
(Not Used)	2 – 6
BIAS TOO LOW Fault (auto bias only)	3 – 1
HV TOO LOW Fault	3 – 2
48V TOO LOW Fault	3 – 3
15V TOO LOW Fault	3 – 4
–15V TOO LOW Fault	3 – 5
SINK TEMP TOO HIGH Fault	3 – 6
DELTA TEMP TOO HIGH Fault	4 – 1
AVG POWER TOO HIGH Overload	4 – 2
BIAS TOO HIGH Fault (auto bias only)	4 – 3
HV TOO HIGH Fault	4 – 4
48V TOO HIGH Fault	4 – 5
15V TOO HIGH Fault	4 – 6
–15V TOO HIGH Fault	5 – 1
EEPROM Fail Fault	5 – 2
ADC Fail Fault	5 – 3

4.4 Gain Adjustment

The amplifier gain has been adjusted to nominal gain at the factory. The gain may be adjusted within a ± 3 dB range by means of the GAIN ADJ control on the front panel. A small flat-bladed screwdriver or a trimpot adjustment tool may be used to perform the adjustment.

5. EVALUATION PROCEDURES

5.1 Required Test Equipment

The tests described require the use of standard RF testing equipment. A complete list of the equipment needed is given below and each test lists the gear required for that operation.

- ☐ Network/Spectrum analyzer (HP4396A or equivalent).
- ☐ Transmission/Reflection test kit (HP87512A).
- ☐ RF signal generator (two needed for IMD).
- ☐ Pulse generator.
- ☐ Signal combiner.
- ☐ 50 dB attenuator.
- ☐ 50 ohm load with 2000 watt minimum power capability.
- ☐ 30 dB amplifier.
- ☐ Cabling and terminations, as required.

These procedures assume a working knowledge of the required test equipment to perform the evaluations.

5.2 General Test Setup

Figure 5-1 displays the general test setup required for system evaluation. Specific changes necessary for any test are provided within each procedure.

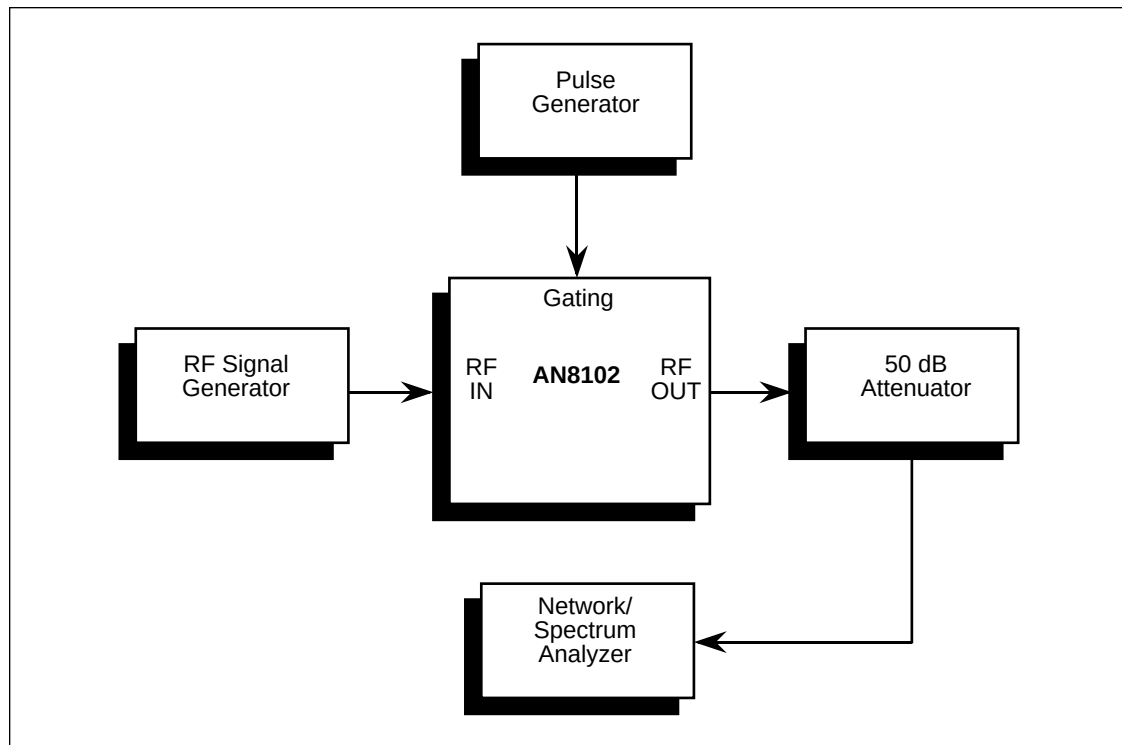


Figure 5-1. General Test Setup.

5.3 Evaluation Test Procedures

5.3.1 Noise Floor

1. In the setup of Figure 5-1, remove the pulse generator cable from the GATING input jack.
2. In the setup of Figure 5-1, remove the signal generator cable from the RF IN connector.
3. In the setup of Figure 5-1, replace the 50 dB attenuator with a low noise 30 dB amplifier.
4. Assert AC ON signal and wait for the READY LED to light.
5. Check the noise level from RF OUT jack as measured on the spectrum analyzer.
6. The noise floor is 30 dB below the measured level on the spectrum analyzer.

5.3.2 Pulse Droop

1. Set up the test equipment as in Figure 5-1.
2. Assert AC ON signal and wait for the READY LED to light.
3. Set the RF signal generator output to -2 dBm at the operating frequency.
4. Provide a GATING signal input within the pulse specifications.
5. Set the spectrum analyzer to zero span at the operating frequency.
6. Adjust RF signal generator output to obtain an initial pulse amplitude of full nominal output.
7. View and measure pulse droop during the active GATING signal.
8. At test completion, remove the GATING signal to disable the RF output.

5.3.3 Gain Linearity

1. Set up the test equipment as in Figure 5-1.
2. Assert AC ON signal and wait for READY LED.
3. Set the RF signal generator output to -2 dBm at the operating frequency.
4. Provide a GATING signal input within the pulse specifications.
5. Set the spectrum analyzer to zero span at the frequency of RF input.
6. Adjust RF signal generator output to obtain an initial pulse amplitude of full nominal output.
7. Record output level on spectrum analyzer.
8. Lower RF IN from signal generator by 1 dB and repeat Step 5. Continue this procedure to -20 dB below the full power input level.
9. Decrease signal generator output from -20 dB to -40 dB below full power in 5 dB intervals. Record the RF OUT level for each interval.
10. Remove the GATING signal to disable the RF power output.
11. Evaluate gain linearity results by noting deviation of data from a straight line output vs. input power curve, as shown in Figure 5-2.

Note: *The above test can be done using the HP4396A Network/Spectrum Analyzer by sweeping the generator output from 0 dBm to -30 dBm and both gain and phase linearity can be measured instantaneously.*

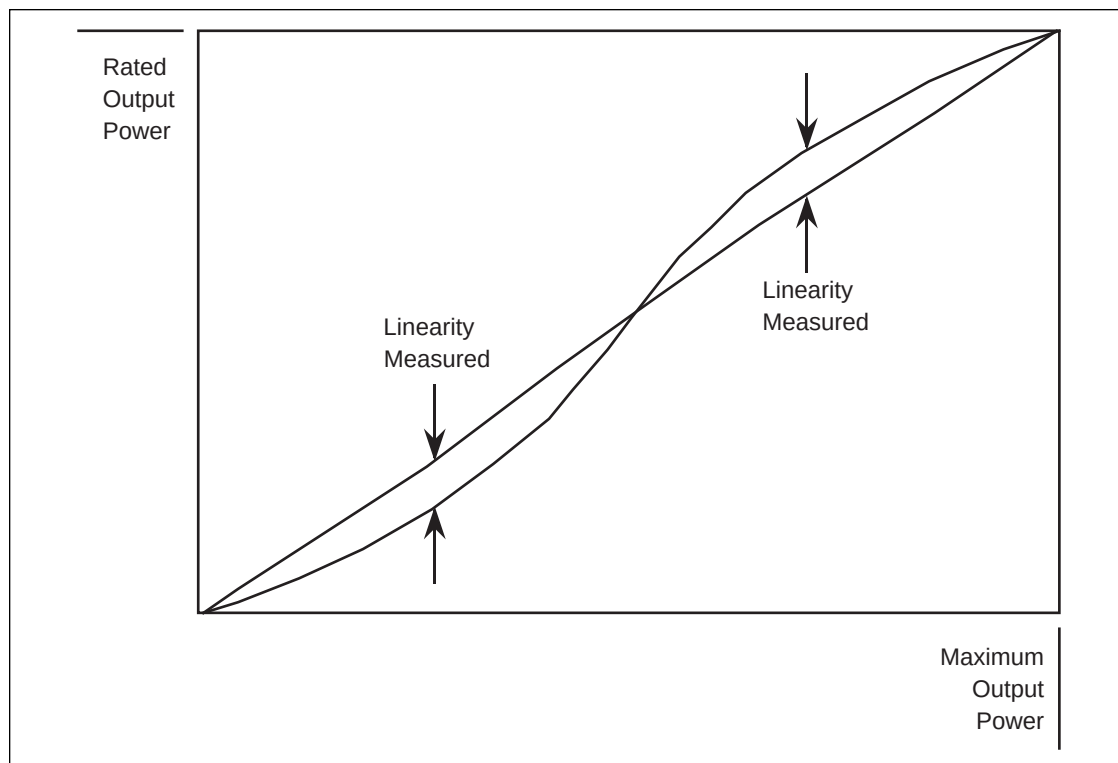


Figure 5-2. Gain Linearity.

5.3.4 Gain Stability

1. Set up the test equipment as in Figure 5-1.
2. Assert AC ON signal and wait for READY LED.
3. Set the RF signal generator output to -2 dBm at the operating frequency.
4. Provide a GATING signal input within the pulse specifications.
5. Set the spectrum analyzer to zero span at the operating frequency.
6. Adjust RF signal generator output to obtain an initial pulse amplitude of full nominal output.
7. On the spectrum analyzer, view RF OUT level.
8. Monitor stability of output power for five minutes and note the maximum variation in output.
9. Remove the GATING signal to disable the RF output.

5.3.5 Frequency Response

1. Set up the test equipment as in Figure 5-1.
2. Assert AC ON and wait for READY LED.
3. Set the Network Analyzer center frequency to the operating frequency and the frequency span to 1MHz, adjust the RF output level to -2 dBm.
4. Provide a GATING signal input within the pulse specifications.
5. Adjust the network analyzer output level to obtain full peak output from the amplifier at the operating frequency.
6. Record output level on the spectrum analyzer.
7. Remove the GATING signal to disable the RF output.

5.3.6 Harmonic Distortion

1. Set up the test equipment as in Figure 5-1.
2. Assert AC ON and wait for READY LED.
3. Set the RF signal generator output to -2 dBm at the operating frequency.
4. Provide a GATING signal input within the pulse specifications.
5. Set the spectrum analyzer to sweep from 10 MHz to 110 MHz.
6. Adjust the RF signal generator to obtain full peak output from the amplifier.
7. Measure the harmonic levels and compare to the test specifications.
8. Remove the GATING signal to disable the RF output.

5.3.7 Intermodulation Distortion

Intermodulation distortion (IMD) requires the use of two signal generators and a combiner as shown in Figure 5-3.

1. Set up the test equipment as in Figure 5-3.
2. Set both signal generators to 0 dBm. Set frequency of one generator to 25 kHz below the operating frequency and the other generator to 25 kHz above the operating frequency to provide 50 kHz tone separation.
3. Provide a GATING signal input within the pulse specifications.
4. Adjust the spectrum analyzer to view output spectrum of amplifier, including the harmonic levels.
5. Evaluate the spectrum against Figure 5-4 to determine distortion levels.
6. Remove the GATING signal to disable the RF output.

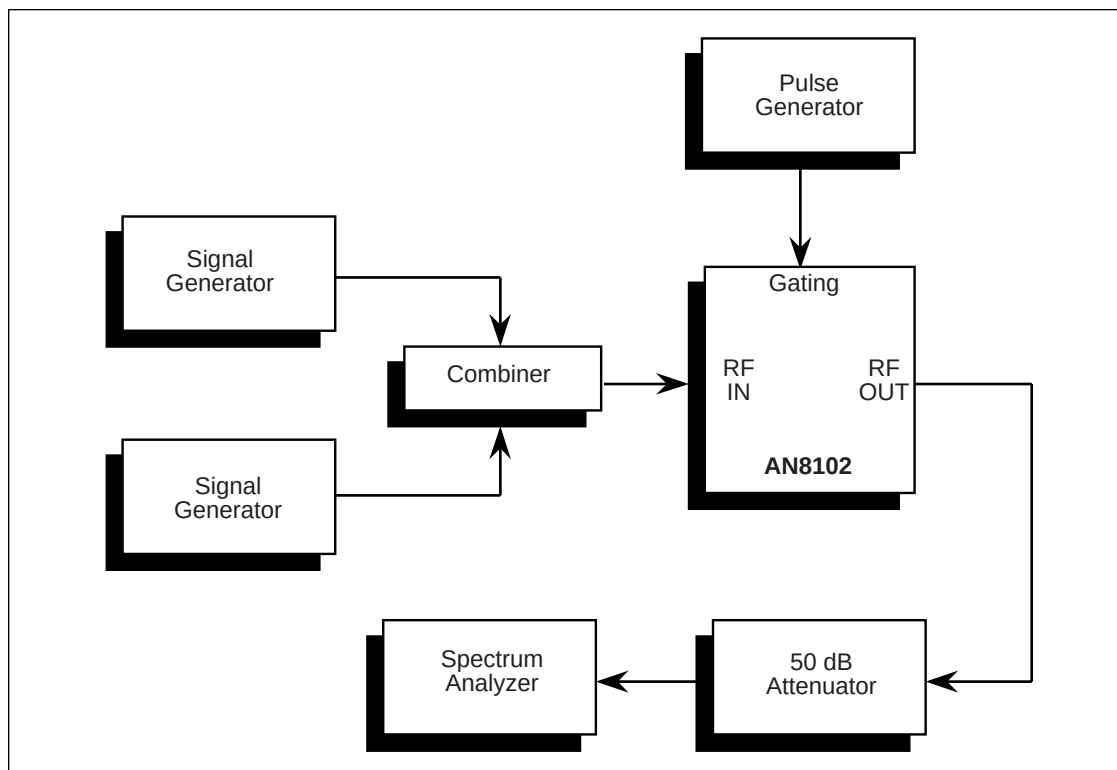


Figure 5-3. IMD Test Setup.

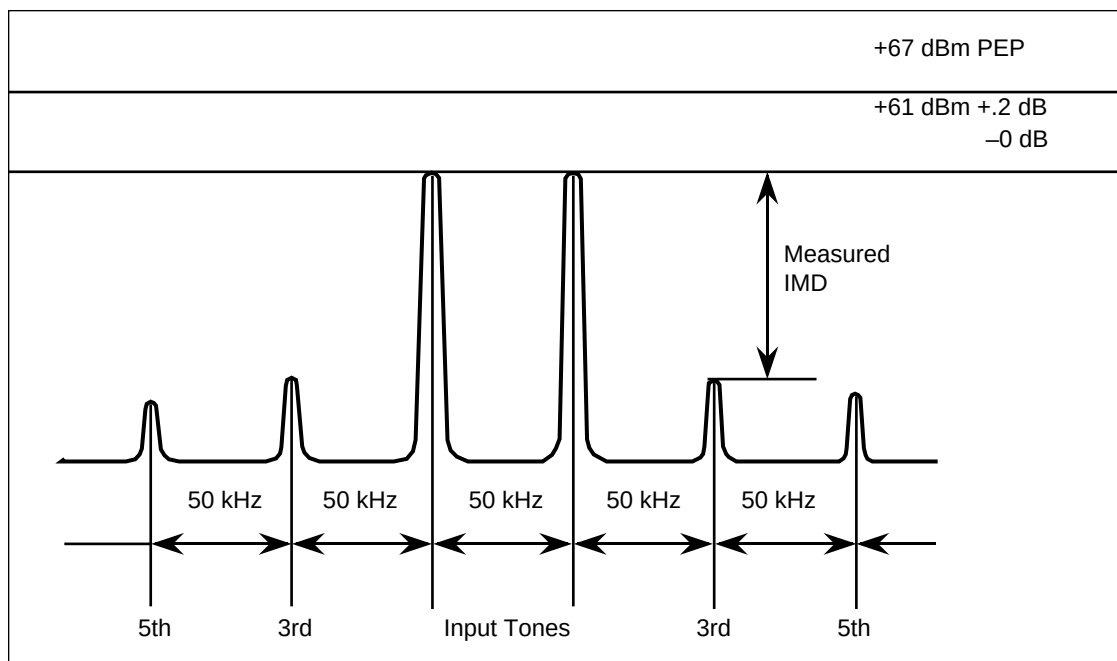


Figure 5-4. IMD Spectrum.

5.3.8 Input Impedance

Determination of input impedance requires the measurement of both Forward and Reflected RF power. .

1. Set up the test equipment as in Figure 5-5.
2. Set the RF signal output from the network analyzer to -2 dBm at the operating frequency $\pm$ 500 kHz .
3. Provide a GATING signal input within the pulse specifications.
5. Adjust RF network analyzer output to obtain an initial pulse amplitude of full nominal output.
6. Measure the Input VSWR using the network analyzer after properly calibrating the network analyzer.

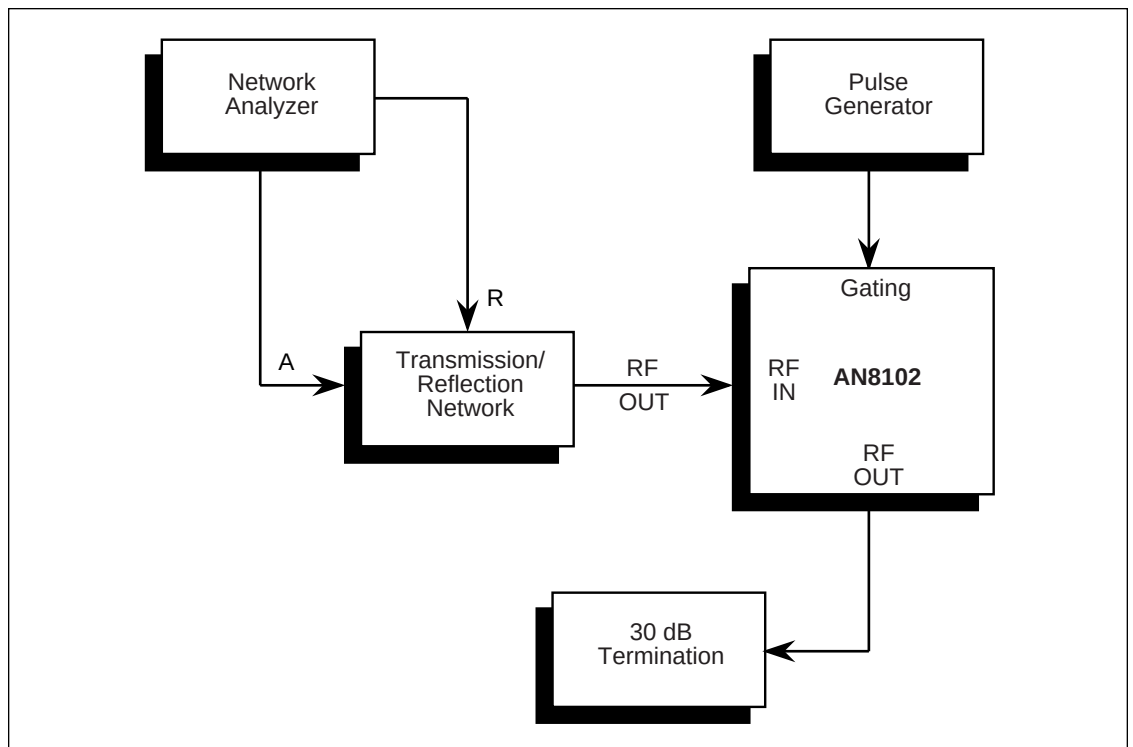


Figure 5-5. Input Impedance Test Setup.

6. MAINTENANCE & TROUBLESHOOTING

Maintenance of the AN8102 is straightforward and simple.

6.1 Preventive Maintenance

General preventive maintenance involves standard procedures used to maintain electrical instrumentation. These procedures should be performed every three months.

- ☐ Inspect the AC line cord for fraying or damage. Replace a questionable power cord immediately.
- ☐ Check the chassis for continuity to earth ground. Poor grounding conditions represent a hazard for personnel who may come into contact with the amplifier.
- ☐ Insure that placement of the amplifier meets the environmental specifications outlined in Section 2.

WARNING



The AN8102 contains high voltages. The system should be disconnected from the main power and all high voltage locations discharged before performing maintenance inside the chassis.

AVERTISSEMENT

Le AN8102 contient de très hautes tensions, avant de faire aucune maintenance à l'intérieur du système, il faut le débrancher de la source principale puis décharger toutes les surfaces à hautes tensions.

WARNUNG

Im AN8102 treten hohe spannungen auf. Vor der durchführung von wartungsarbeiten im gehäuseinneren sollten der netzanschluss getrennt und alle hochspannungspunkte entladen werden.

6.2 Troubleshooting Information

When an error is detected within the amplifier system, the diagnostic logic attempts to classify the error as an overload or as a fault.

Overloads are those errors caused by EXCESSIVE user input (i.e. excessive duty cycle, RF input overdrive, too much reflected power, etc.). An overload condition puts the amplifier in a self-protection “off” state. Removing the error condition and resetting the amplifier will restore normal operation.

Faults are error conditions most likely caused by an internal circuit failure (i.e. blown transistor, etc.). True faults permanently disable the amplifier.

The status section of front panel indicators displays the state of the amplifier as described below

<u>POWER</u>	<u>READY</u>	<u>FAULT</u>	<u>AMPLIFIER CONDITION</u>
OFF	OFF	OFF	No AC mains power to the Amplifier, or main circuit breaker off, or blown STDBY power supply (unlikely). If so replace the unit.
ON	OFF	OFF	Unit in standby mode (no AC on command) or AC-on cable not plugged in.
ON	ON	OFF	Unit ready mode.
ON	OFF	ON	Unit in fault/overload condition. Check diagnostic code for specific condition.

6.3 Fault Diagnostics

Suspected hardware failures may be diagnosed using tools given in this section. Individual printed circuit boards within the AN8102 are not meant to be diagnosed to the component level by the user. Inoperable circuit boards should be returned to ANALOGIC for replacement.

Section 2, part 2.11 shows a description of status indicators on the front panel.

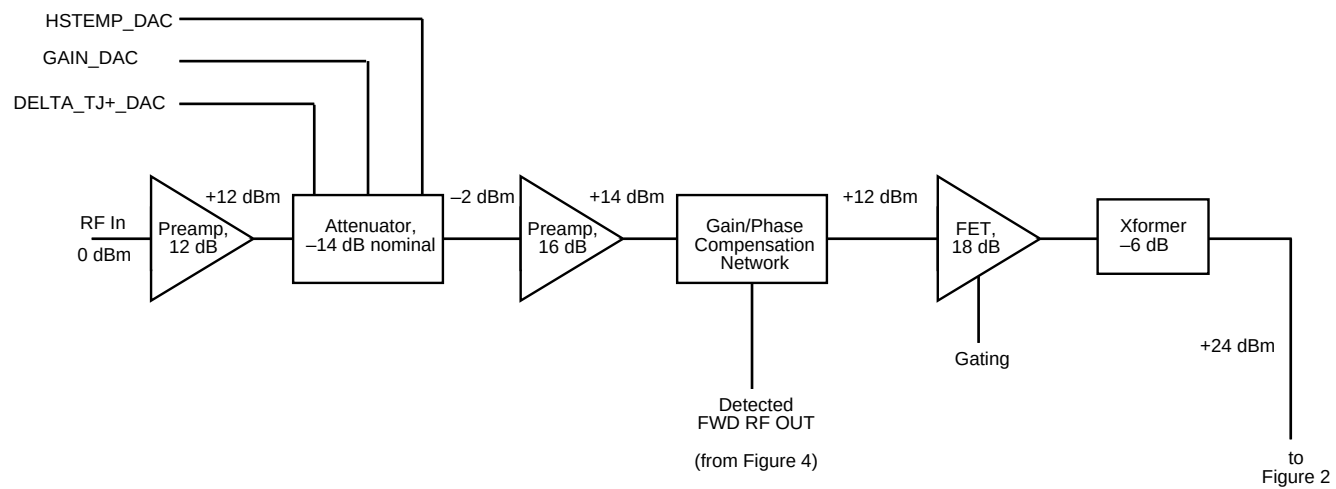
Fault and Overload conditions (FAULT light on) are accompanied by a series of flashes of READY light. Refer to **Section 4.3.3** for more details.

Also, the failure type will be reported on the diagnostic link by MODE command. The following table lists the set of Overload/Fault messages that can be received from the communication link through the 15-pin “D” connector. Refer to **Section 3.3** for pin numbers used for communication through this connector.

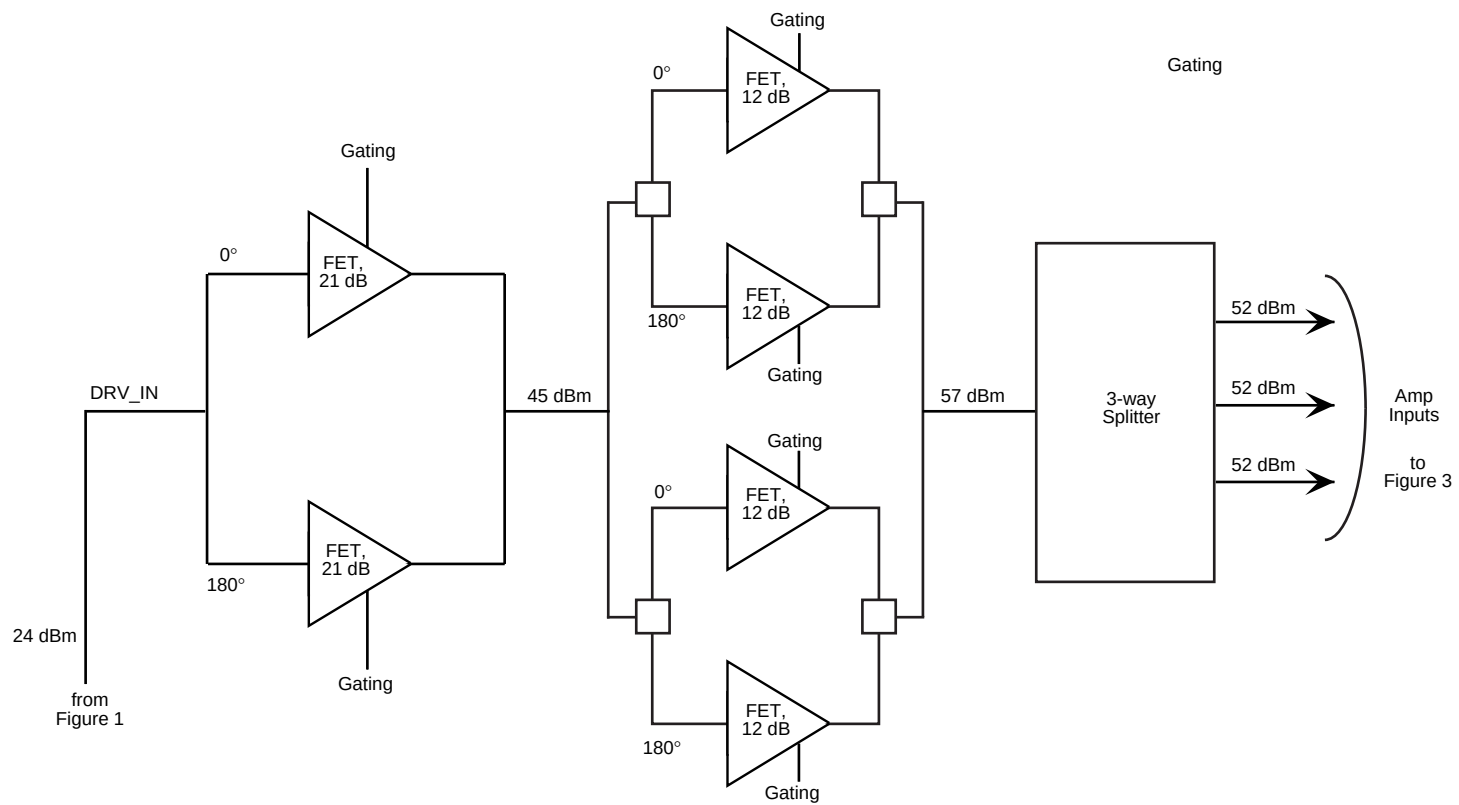
Appendix A

List of Drawings

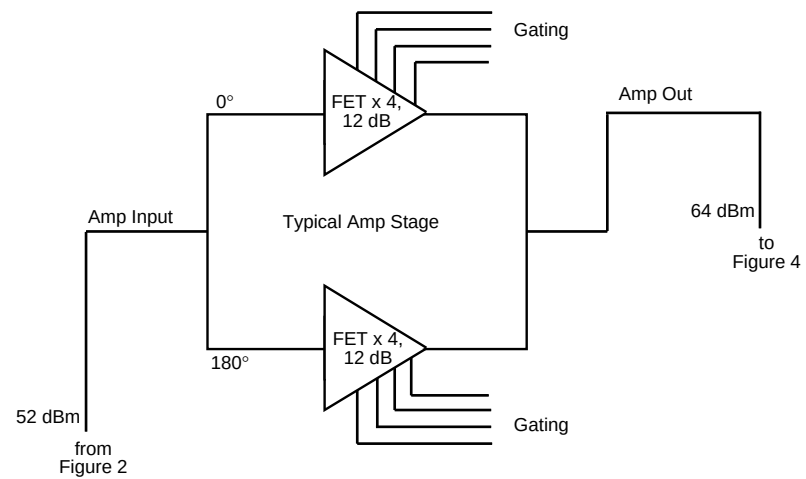
Drawing #	Title
9-26891	Family Tree AN8102 9.0 kW RF Amplifier
5-41539	System Wiring Diagram AN8102
AN8102	Functional Block Diagram (4 pages)



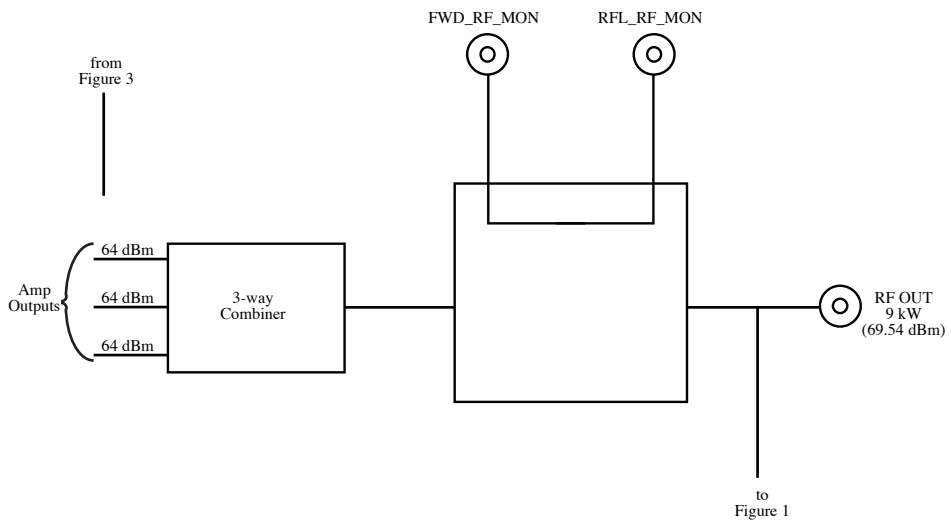
AN8102 Functional Block Diagram, Figure 1 of 4



AN8102 Functional Block Diagram, Figure 2 of 4



AN8102 Functional Block Diagram, Figure 3 of 4



AN8102 Functional Block Diagram, Figure 4 of 4

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